DPP — DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CARE OF NEWBORN WITH NEONATAL SEPSIS
DEPARTMENT:	NEONATAL INTENSIVE CARE UNIT
APPLIES TO:	NICU PHYSICIANS AND NURSES

CODE AND VERSION NUMBER	
DPP-NICU-0017 (v4)	
LAST VERSION APPROVAL DATE:	22 / 05 / 2023
NUMBER OF PAGES:	7

	NAME	POSITION	SIGNATURE
PREPARED BY:	MS. KRISTINE P. MAGNETICO	UNIT QUALITY REPRESENTATIVE	
	MS. SHATHA MOHAMMED	NICU HEAD NURSE	
REVIEWED BY:	MEDICAL EXECUTIVE COMMITTEE		
	DR. ABDULKARIM IBRAHIM	HEAD OF NICU	
	MS. MERRY JOY H. GALLEBO	NURSING QUALITY COORDINATOR	
	MR. KHALID GHURMULLAH ALGHAMDI	QPS DIRECTOR	
APPROVED BY:	MS. HAIFA SAAD MISFER	DIRECTOR OF NURSING	
	DR. RAKAN KHALIL ALGHAMDI	AED FOR MEDICAL SERVICES	

APPROVED DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
09 / 02 / 2026	23 / 02 / 2026	TRIENNIAL	08 / 02 / 2029

CONTENTS

1. PURPOSE	4. POLICY	7. REFERENCE/CROSS-REFERENCES
2. DEFINITION & ABBREVIATION	5. PROCEDURE	8. ATTACHMENT
3. RESPONSIBILITY	6. MEASUREMENT	9. REVISION CLAUSE

1. PURPOSE

1.1. To provide guidelines to health care providers on how to care for newborn with neonatal sepsis.

1.2. To promote prompt and effective treatment for neonates with suspected and confirmed sepsis.

1.3. To prevent or reduce incidence of late-onset sepsis.

1.4. To ensure awareness of staff regarding the risk and complications of neonatal sepsis.

2. DEFINITION & ABBREVIATION

2.1. **Neonatal Sepsis** – is defined as a clinical syndrome of infection which gain access into the blood stream with systemic signs and symptoms exhibited in the first four (4) weeks of life.

2.2. **Early-onset sepsis** – an infection mostly associated with acquisition of microorganisms from the mother or during delivery in the first 48-72 hours.

2.3. **Late-onset sepsis** – is caused by the organisms thriving in the external environment of the home or the hospital and often transmitted through the hands of the healthcare providers that occur after 72 hours of age.

2.4. **Blood Culture** – is a test taken to look for pathogens that are causing the infection in the bloodstream that interferes with the newborn body's normal defenses.

2.5. **Sclerema Neonatorum** – is a rare and severe skin condition that is characterize by diffuse hardening of the subcutaneous of the subcutaneous tissue with minimal inflammation.

2.6. **STAT (Statim)** – immediately or at once

2.7. **VLBW** –Very Low Birth Weight

2.8. **CBC** –Complete Blood Count

2.9. **ESR** – Erythrocyte Sedimentation Rate

2.10. **CRP** – C-Reactive Protein

2.11. **CRT** –Capillary Refill Time

2.12. **NEC** –Necrotizing Enterocolitis

2.13. **PROM** – Premature Rupture of Membrane

2.14. **PPE** – Personal Protective Equipment

3. RESPONSIBILITY

3.1. **Head of Neonatology Department**

3.1.1. Responsible of ensuring that all clinical practices are applied by all medical and nursing staff to prevent nosocomial sepsis in the unit.

3.2. **Physicians (Consultant, Specialists and Residents)**

3.2.1. Responsible to assess and evaluate newborn with clinical indicators suggestive of possible sepsis.

3.2.2. Write the orders of the procedure to treat and manage neonatal sepsis in the medical file.

3.3. **NICU Nurse (Head Nurse, Charge Nurse and Staff Nurse)**

3.3.1. Will assess and monitor patient with neonatal sepsis.

3.3.2. Do verify, carry out and document any procedures done to patient in the medical file.

4. POLICY

4.1. All newborn patients must be carefully assessed and examined for possible and confirmed sepsis and will be subjected for admission to Neonatal Intensive Care Unit according to the newborn's clinical condition and physician's written order.

4.2. Hand hygiene and wearing of PPE must be strictly implemented to control the spread of infection in the Neonatal Intensive Care Unit. Refer to MPP-0032 IPC Guidelines in NICU.

4.3. Medical devices in NICU such as laryngoscope blades, incubators, monitoring devices that are used in patient care must be cleaned and disinfected according to IPC protocols. Refer to MPP-IPC-0030 IPC Guidelines in Cleaning and Disinfection of Medical Equipment.

4.4. The unit should implement the control of visitors to avoid overcrowding in the unit. Refer to MPP-NICU-0001 Visiting Hours.

4.5. **Assessment criteria of newborn for sepsis**

4.5.1. Risk factors: Early onset sepsis (1st 72 hours)

4.5.1.1. Prolonged rupture of membranes (greater than 18 hours)

4.5.1.2. Prematurity (especially if associated with PROM)

4.5.1.3. Preterm labor without adequate reason

4.5.1.4. Fetal distress without adequate reason (fetal heart abnormalities especially fetal tachycardia)

4.5.1.5. Maternal fever or other evidence of infection

4.5.1.6. Foul-smelling amniotic fluid or malodorous baby

4.5.1.7. Multiple per vaginum examinations

4.5.1.8. Prolonged labor and aspiration of meconium

4.5.2. Risk factors: Late onset (after 72 hours)

4.5.2.1. Prematurity and low birth weight

4.5.2.2. VLBW infants indwelling catheters, central lines and chest drains

4.5.2.3. Lack of breastfeeding

4.5.2.4. Superficial infections (pyoderma, umbilical sepsis)

4.5.2.5. Aspiration from feeds

4.5.2.6. Disruption of skin integrity from needle pricks and use of intravenous fluids

4.6. **Clinical Manifestations of Neonatal Sepsis**

4.6.1. Respiratory: cyanosis, tachypnea, grunting, apnea, gasping, chest retractions

4.6.2. CNS: lethargy, hypoactive, seizures, irritable

4.6.3. Gastrointestinal: poor sucking reflex, abdominal distention, vomiting, diarrhea

4.6.4. Cardiovascular: bradycardia, bleeding, shock, prolonged capillary refill time

4.6.5. Body temperature changes

4.6.6. Jaundice

4.6.7. hypoglycemia

4.7. Chest radiography must be done and ordered by the physician.

4.8. Urine culture and analysis by suprapubic aspirate or catheter should be done.

4.9. Laboratory test must be done to diagnose neonatal sepsis and identify the bacteria that are causing the infection, which includes extracting blood culture, C-reactive protein and CBC samples.

4.10. A positive "sepsis screen" takes into account of two (2) or more positive test as given below:

4.10.1. Leukopenia (TLC less than 5,000/cmm)

4.10.2. Neutropenia (ANC less than 1800/cmm)

4.10.3. Immature neutrophil to total neutrophil (I/T) (greater than 0.2) which means that immature neutrophils are over 20 percent of the total neutrophils because bone marrow pushes even the premature cells into circulation to fight infection.

4.10.4. Platelet count of less than 100,000 per cmm.

4.10.5. Toxic granules on peripheral smear

4.10.6. Gastric aspirates smears showing more than 5 leukocytes

4.10.7. Micro ESR (greater than 15 in 1st hour)

4.10.8. CRP positive which has a high degree of sensitivity for neonatal sepsis

4.11. Rechecking of CRP must be done after 24-72 hours after commencing antibiotics for those newborn with initial positive CRP results.

4.12. Lumbar puncture will be done if there is strong suspicion of sepsis, if CRP above 10mg/L, if with clinical signs of meningitis or upon the physician's discretion to examine the cerebrospinal fluid for bacteria.

4.13. Cardiopulmonary status must be stabilized with the infusion of normal saline (10ml/kg over 5-10 minutes, if perfusion is poor as evidenced by capillary refill time of more than 3 seconds.

Repeat the same dose 1-2 times over the next 30-45 minutes, if perfusion continues to be poor.

4.14. Correct hypoglycemia, infuse glucose (10 percent) 2ml/kg stat.

4.15. **Antibiotic therapy**

4.15.1. A combination of ampicillin and gentamicin is recommended for treatment of sepsis and pneumonia. In cases of suspected meningitis, cefotaxime should be used along with aminoglycosides.

4.15.2. Review the clinical progress and microbiology reports at 48 hours

4.15.3. Metronidazole can be added if there is suspicion of anaerobic infection (e.g. intra-abdominal sepsis, NEC)

4.15.4. Consider vancomycin for coagulate negative staphylococcal sepsis especially if patient is unwell or with central line infection.

4.15.5. In late onset sepsis, a first line antibiotic is cloxacillin (100mg/kg/day) and aminoglycoside. In nosocomial sepsis, if antibiotics sensitivity pattern of organisms is known then start antibiotics accordingly.

4.15.6. Superficial infections must be adequately managed since it can lead to sepsis when neglected but can be treated with local application of antimicrobial agents.

4.15.7. Treating for a minimum of 7-day course of intravenous antibiotics is suggested where there is strong suspicion of sepsis or a positive blood culture but this may vary depending on the pathogen and the clinical status of the baby.

4.15.8. Consider stopping the antibiotics after 36 hours if the blood culture is negative, the newborn is clinically stable, the suspicion of infection was low and if CRP trend is reassuring, otherwise continue therapy if culture is positive or sepsis is very likely.

4.16. **Supportive care for the septic neonate**

4.16.1. Provide warmth and ensure and ensure consistently normal temperature nursed in a thermoneutral environment.

4.16.2. Prevent bleeding by administering Vitamin K 1 mg intramuscularly.

4.16.3. Consider the use of dopamine if perfusion is persistently poor.

4.16.4. Avoid enteral feed if newborn is very sick but provide maintenance fluids intravenously.

4.16.5. Consider performing exchange transfusion with fresh whole blood if there is scleremaneonatorum.

5. PROCEDURE

5.1. The physician and nurse will assess the patient for the clinical manifestation of neonatal

sepsis.

5.2. The physician will write the order of the procedure in the medical file to treat/manage sepsis.

5.3. Verify and carry out the physician's order.

5.4. Identify the patient correctly.

5.5. Do hand hygiene and wear appropriate PPE according to the procedure to be done.

5.6. Perform neonatal sepsis management according to physician's order.

5.6.1. Observation and monitoring of vital signs

5.6.2. Blood investigations

5.6.3. Antibiotic therapy

5.7. Perform continuous re-assessment and follow-up blood culture and CRP report of the sick newborn.

5.8. The physician will change the line of antibiotics accordingly, if blood culture and CRP are positive.

5.9. The physician and nurse will continue to re-evaluate the newborn's condition.

5.10. If the newborn remains vitally and clinically stable, stop antibiotic therapy if blood culture and CRP reports are negative, according to physician's written order.

5.11. The nurse will provide the appropriate nursing care for the septic newborn according to physician's ordered treatment and management of sepsis.

5.12. Document all the procedure done to patient in the medical file.

6. MEASUREMENT

6.1. 100% of the staff is knowledgeable and adhere to this policy and procedures.

7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0017(v3): Care of Newborn with Neonatal Sepsis, 22/05/2023, p. 1-6

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10 p.126-127

7.3. Neonatal Sepsis and Meningitis, Philip AGS, GK Hall Medical Publishers

7.4. National Institute for Health & Clinical Excellence UK

7.5. http://www.umm.edu/health/ency/articles/neonatal-sepsis

8. ATTACHMENT

8.1. None

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster

logo, font face and design.